NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

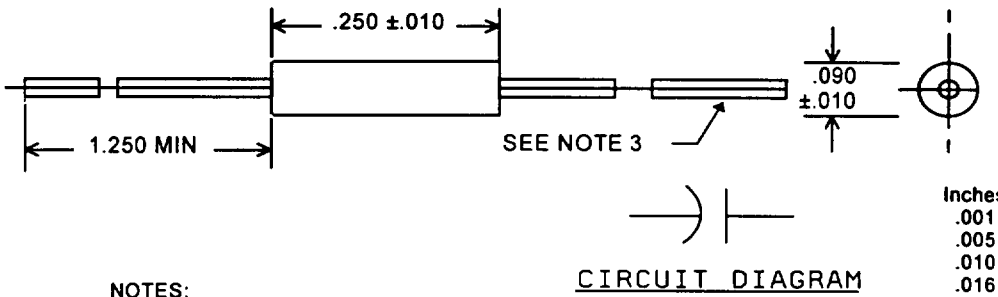
MIL-PRF-20/28B
9 May 1989
SUPERSEDING
MIL-C-20/28A
5 November 1976

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR76 AND CC76

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-20.



CIRCUIT DIAGRAM

Inches	mm
.001	0.03
.005	0.13
.010	0.25
.016	.041
.090	2.29
.160	4.06
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .015 (0.38 mm) to .027 (0.69 mm).
4. Ⓑ Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Styles CCR76 and CC76 capacitors.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR76 only): M (1.0%), P (0.1%), R (.01%), or S (.001%).

Ⓑ denotes changes

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Thermal shock and voltage conditioning (CCR76 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

TABLE I. Capacitor characteristics.

Type 1/ 2/ designation	Rated voltage	Nom. cap.	Cap. tol.	Type 1/ 2/ designation	Rated voltage	Nom. cap.	Cap. tol.
	Volts, dc	pF			Volts, dc	pF	
CC-76CG820--	200	82	FGJ	CC-76CG431--	100	430	FGJ
CC-76CG910--	200	91	FGJ	CC-76CG471--	100	470	FGJ
CC-76CG101--	200	100	FGJ	CC-76CG511--	100	510	FGJ
CC-76CG111--	200	110	FGJ	CC-76CG561--	100	560	FGJ
CC-76CG121--	200	120	FGJ	CC-76CG621--	100	620	FGJ
CC-76CG131--	200	130	FGJ	CC-76CG681--	100	680	FGJ
CC-76CG271--	100	270	FGJ	CC-76CG751--	50	750	FGJ
CC-76CG301--	100	300	FGJ	CC-76CG821--	50	820	FGJ
CC-76CG331--	100	330	FGJ	CC-76CG911--	50	910	FGJ
CC-76CG361--	100	360	FGJ	CC-76CG102--	50	1000	FGJ
CC-76CG391--	100	390	FGJ				

1/ Complete type designation will include the following:
First dash: Symbol "R" (for style CCR76) or dash will be deleted (for style CC76).
Second dash: Applicable capacitance tolerance symbol.
Third dash: Applicable failure rate level symbol (CCR76 only) or dash will be deleted (for style CC76).

2/ Capacitors deleted since the previous issue shall be replaced by selecting a capacitor with the nearest parameter.

Marking: Method I of MIL-STD-1285, as shown in following example:

(B) JCG - "J" brand (ER only) (J); and characteristic (CG).
820 - Capacitance value.
GM6 - Capacitance tolerance (G); FR level, CCR76 only (M); and year code (9 for 1989).
ATM - Lot code (A); and trademark (TM).

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
DLA - ES

User activities:
Navy - AS, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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